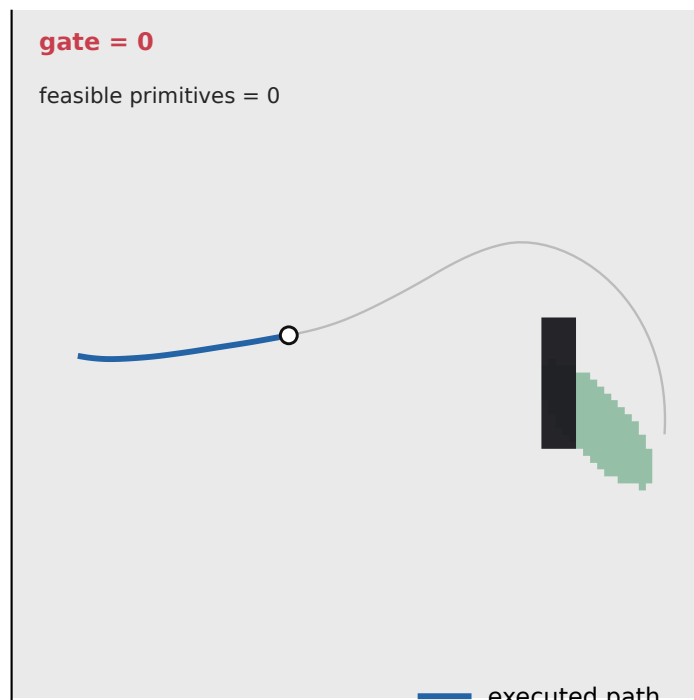


RQ1 — The witness gate suppresses unnecessary soft-force exposure

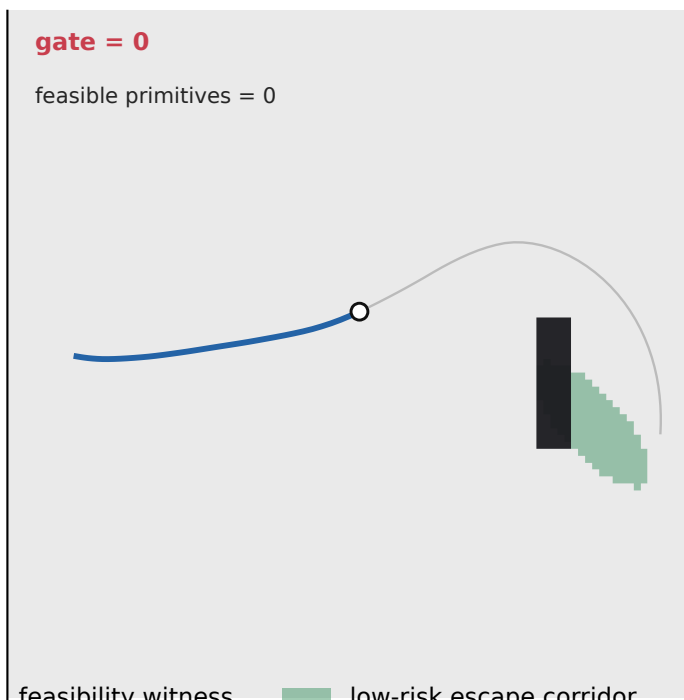
A

Nominal route
step 31



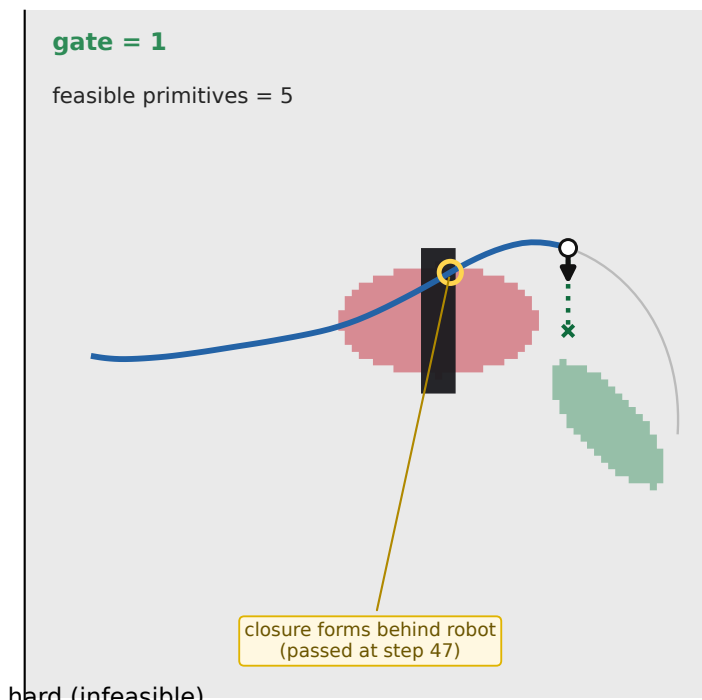
B

Escape visible but blocked
step 39



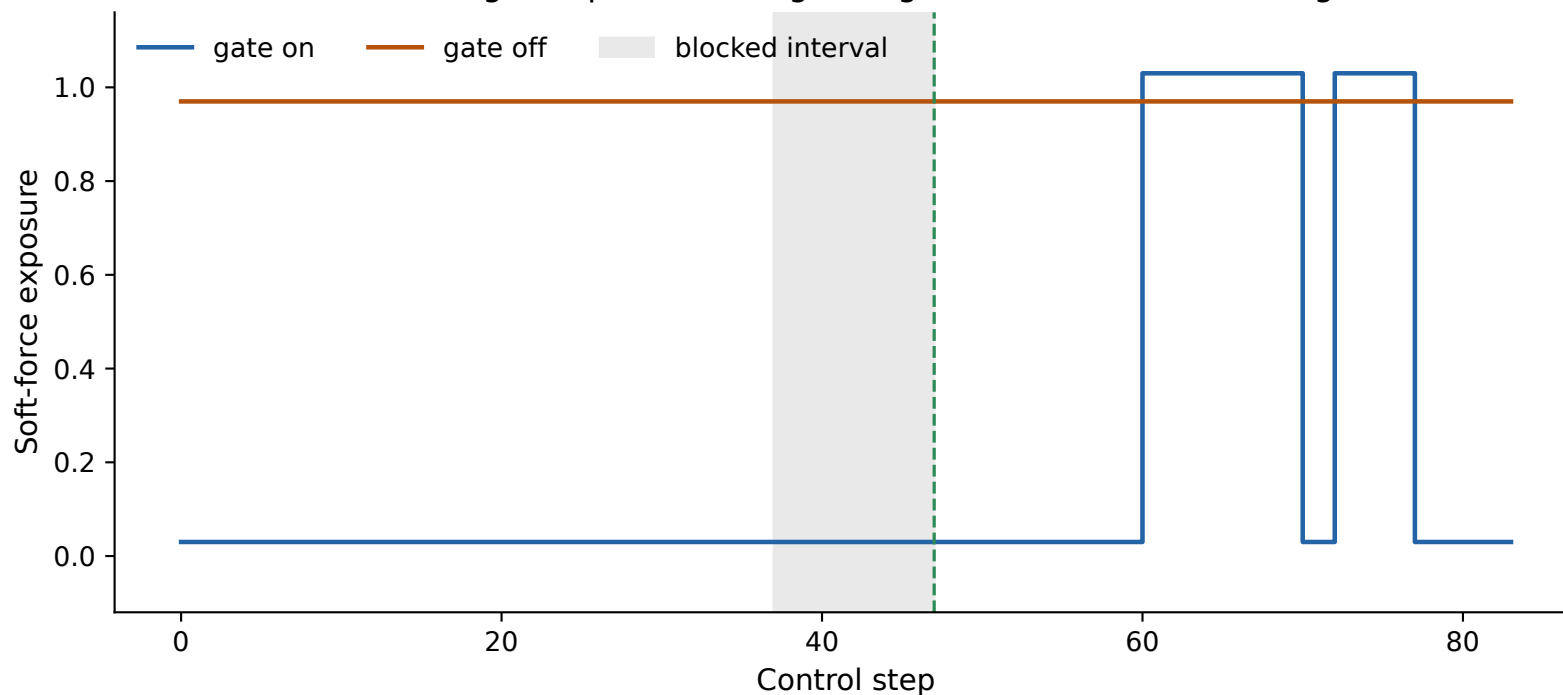
C

Escape feasible: gate activates
step 60



D

Gate changes exposure timing; navigation outcome is unchanged



E

Exposure falls by 91--94%

